

Validation report

Model version 1.0.0, built 2026-10-08. Fit on 2014–2023; scored on held-out seasons 2024, 2025 (569 games). Splits are by season, so plays from one game never sit on both sides.

Win probability (every held-out snap)

Model	Plays	Log loss	Brier
This engine	81,654	0.446	0.148
nflfastR <code>vegas_wp</code> (same plays; nflfastR was trained on them)	81,654	0.4457	0.1475
nflfastR <code>wp</code> without the spread	81,654	0.4841	0.1641
Score-and-time-only logistic baseline	81,654	0.4931	0.1672
This engine, final 5 minutes only	9,427	0.2501	0.0812

Calibration (predicted vs actual win rate by predicted-probability bin):

Bin	Plays	Predicted	Actual
0.00-0.10	11,790	3.4%	3.5%
0.10-0.20	7,786	15.0%	16.6%
0.20-0.30	7,226	24.8%	26.1%
0.30-0.40	6,755	35.0%	34.3%
0.40-0.50	6,271	44.9%	43.4%
0.50-0.60	6,202	55.1%	55.2%
0.60-0.70	7,068	65.1%	64.4%
0.70-0.80	7,608	75.1%	73.6%
0.80-0.90	7,889	85.0%	83.5%
0.90-1.00	13,059	96.4%	96.1%

Option values (the test that matters for decisions)

For each held-out 4th down, the engine's win probability for the option the coach actually chose, against how often that team won:

Coach chose	Plays	Engine predicted	Actual win rate
go	670	37.3%	38.1%
fg	826	56.3%	55.6%
punt	1501	47.2%	46.6%

Conversion on 4th down

Log loss 0.6424 (n = 1849); with team strength 0.6406.

Yards to go	Plays	Predicted	Actual
1	721	69.7%	71.3%
2-3	491	60.9%	60.3%
4-6	332	51.3%	56.3%
7-10	179	37.3%	36.9%
11-99	126	21.8%	32.5%

Field goals

Factors	Kicks	Log loss	Brier
base	2306	0.3688	0.1147
weather	2306	0.3671	0.114
stadium	2306	0.3676	0.1142
weather+stadium	2306	0.3673	0.114

Distance	Kicks	Predicted	Actual
18-29	469	98.0%	97.7%
30-39	632	93.3%	93.7%
40-49	640	81.8%	80.8%
50-54	354	69.3%	74.0%
55-70	211	59.3%	61.1%

Kicking-era term chosen on the hold-out: **trend** (log loss: trend 0.3688, era2020 0.3694, both 0.3697).

Kicker skill priors chosen on the hold-out: accuracy sd 0.4, range sd 0.1. Log loss 0.3673 with no kicker effect, 0.3665 with it (accuracy only: 0.3666).

Punts

Mean absolute error of the predicted opponent start: **6.75 yd** (n = 3,714), against 10.9 yd for a league-average guess.

Agreement with coaches (descriptive)

On 3,000 sampled held-out 4th downs the engine's call matched the coach's 62.4% of the time. Where the engine favored going by 2+ points (694 plays), coaches went 47.1% of the time.

Clock and rules measured from the data

Event	Seconds to next snap (normal)	(final 2:00 of half)
go_success	37.0	18.0
go_fail	5.0	6.0
punt	9.0	9.0
fg	5.0	5.0
score	8.0	6.0
play_first	37.0	11.0
play_short	36.0	6.0
punt_fair_catch	7.0	7.0
timeout_play	—	6.0
spike_total	—	16.0

Kickoff: the receiving team starts at yardline_100 **69.3** on average under the 2025 rules. Punt return TD rate 0.39%; muff recovered by the kicking team 1.44%.

Weather: outdoor wind or temperature imputed on 1,425 plays after parsing the weather text.

What these numbers do not prove

- That following the engine wins more games. Option values are calibrated on average, which is necessary, not sufficient. No randomized test of 4th-down policy exists.
- That any single call is right. See the bootstrap confidence for that call.
- Anything about seasons with different rules. The kickoff spot is measured from one rule season.
- That short-yardage conversion is unbiased. Yards to go is a whole number (Lopez 2020; risk 1).